Video 11.1.1

There are two fundamental concepts in the theory of subspaces:

1. Span of a set of vectors.
2. Linear independence of a set of vectors.

Today we meet the second of these:

Linear independence

Some motivation. It's all interesting but why do we ***need*** *this concept?*

One of our goals is to give a precise logical definition of the *dimension* of a subspace.

We want to say:

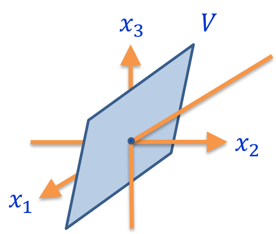
- The subset with just the origin, $\{\mathbf{0}\}$, has *dimension* 0.
- A line through the origin is a subspace of *dimension* 1.
- A plane through the origin is a subspace of *dimension* 2.
- etc.

So how could we define dimension?

First attempt to define dimension:

1. Consider a subspace $V \subset \mathbb{R}^n$.
2. Find a spanning set for V. In other words, find a set S such that $V = span(S)$.
3. Define $\dim(V) = (\#\ of\ elements\ in\ S.)$

Example:



If this subspace V equals

$$span\{(0,1,1),(-1,-1,0)\}$$

then we would say

$$\dim(V) = 2$$

because there are two vectors in the spanning set.

This is almost a good definition. But there is still a major problem with it.

The problem with this definition: A subspace V can have many different spanning sets and the number of vectors in the spanning set might change for different spanning sets. Which spanning set should we count?

Example:

Here are two spanning sets for the same subspace (we can easily check they are the same subspace. *How?*):

$$span\{(1,1,0,0),(1,0,1,0)\} = span\{(1,1,0,0),(1,0,1,0),(2,1,1,0)\}$$

This spanning set has two vectors…

…whereas this spanning set has three vectors.

What's going on?

Example:

Here are two spanning sets for the same subspace:

$$span\{(1,1,0,0),(1,0,1,0)\} = span\{(1,1,0,0),(1,0,1,0),(2,1,1,0)\}$$

… whereas this spanning set has no "redundant" vectors. ***We should count this set to define dimension.***

This spanning set includes a "redundant" vector because $(2,1,1,0) = (1,1,0,0) + (1,0,1,0)$

The concept of **linear independence** is a mathematically logical way to make precise this idea of "having no redundant vectors".

Definition ('Linear dependence').

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

This list of vectors is said to be **linearly dependent** if one of the vectors on the list is a linear combination of the other vectors on the list. $\Rightarrow$ has redundant vectors

Example:

Consider the list: $(1,1,0,0),(1,0,1,0),(2,1,1,0)$.
Note that

$$(2,1,1,0) = (+1)(1,1,0,0) + (+1)(1,0,1,0).$$ or $(1,1,0,0) = (2,1,1,0) + (-1)(1,0,1,0)$

Thus: this list is *linearly dependent*.

Definition ('Linear dependence').

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

This list of vectors is said to be **linearly dependent** if one of the vectors on the list is a linear combination of the other vectors on the list.

Comment:
In this definition I use the word "list" instead of "set" because I want to clearly allow the logical possibility that two of the vectors on the list are the same.

Note: such a list with a repeating vector is definitely linearly dependent. This distinction makes no difference in the case of the definition of span.

The following is an alternative definition for "linearly dependent" which is just a rearrangement of the logic. This is the version you would probably see in textbooks.

Definition ('Linear dependence.' Alternative definition.)

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

This list of vectors is said to be **linearly dependent** if the zero vector can be expressed as a linear combination of the vectors on the list

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m$$

with at least one of these coefficients non-zero: $c_i \neq 0$ for some $1 \leq i \leq m$.

$(+1)(1,1,0,0) + (+1)(1,0,1,0) + (-1)(2,1,1,0)$
$= (0,0,0,0)$

Definition ('Linear dependence.' Alternative definition.)

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

This list of vectors is said to be **linearly dependent** if the zero vector can be expressed as a linear combination of the vectors on the list

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m$$

with at least one of these coefficients non-zero: $c_i \neq 0$ for some $1 \leq i \leq m$.

*This part of the definition is an important detail. We can **always** build the zero vector $\vec{0}$ by*

$$\vec{0} = 0\vec{v}_1 + \cdots + 0\vec{v}_m.$$

But to be "linearly dependent" you need some other way with at least one of the coefficients non-zero.

Video 11.1.2

Mathematicians prefer this version of the definition because it is more "balanced". It treats all the vectors on an equal footing.

Exercise:

Prove that these two definitions are equivalent.

Now that we have defined linear dependence we can define its opposite:

A list of vectors which is not linearly dependent is said to be **linearly independent.**

It will be convenient to write this down as an explicit definition for this shortly.

Exercise:

Prove that the two definitions of linear dependence are equivalent.

Solution:

To prove that two definitions are equivalent, the cleanest approach is to show two things:

1. the first definition implies the second definition.
2. the second definition implies the first definition.

Solution (continued):

Step 1: Proving that the first definition implies the second.

So let $\vec{v}_1, \ldots, \vec{v}_m$ be a list of vectors satisfying the first definition of linear dependence. To be precise, this means that one of the vectors, say $\vec{v}_i$, is a linear combination of the other vectors. In equations we might write:

$$\vec{v}_i = c_1\vec{v}_1 + \cdots + c_{i-1}\vec{v}_{i-1} + c_{i+1}\vec{v}_{i+1} + \cdots + c_m\vec{v}_m.$$

skip Vi!!

Notice that we have skipped the $\vec{v}_i$ term because the definition says "a linear combination of the ***other*** vectors".

Solution (continued):

Step 1: Proving that the first definition implies the second (cont).

Why does this list also satisfy the second definition?

It's very simple. Just take the $\vec{v}_i$ to the other side of the equation. (To be precise: do this by adding $(-1)\vec{v}_i$ to both sides of the equation.)

We deduce:

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_{i-1}\vec{v}_{i-1} + (-1)\vec{v}_i + c_{i+1}\vec{v}_{i+1} + \cdots + c_m\vec{v}_m.$$

* need to check at least one coef. is non-zero

Thus:

- the zero vector $\vec{0}$ is a linear combination of the vectors on the list $\vec{v}_1, \ldots, \vec{v}_m$
- and we are sure that at least one of the coefficients $c_i \neq 0$ because it is (-1).

This is exactly the second definition.

Solution (continued):

Step 2: Proving that the second definition implies the first.

What are we assuming?

There exists an equation amongst the vectors on the list

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m$$

and we know that for at least one of these coefficients, say c_i, that $c_i \neq 0$.

Proof this list satisfies the first definition:

Assume that $c_i \neq 0$ in this equation for some $1 \leq i \leq m$.

Add $(-c_i)\vec{v}_i$ to both sides of this equation. We get:

$$(-c_i)\vec{v}_i = c_1\vec{v}_1 + \cdots + c_{i-1}\vec{v}_{i-1} + c_{i+1}\vec{v}_{i+1} + \cdots + c_m\vec{v}_m$$

Multiply both sides by $-1/c_i$. Because we know $c_i \neq 0$ we are sure we are not dividing by zero. We deduce:

$$\vec{v}_i = (-c_1/c_i)\vec{v}_1 + \cdots + (-c_{i-1}/c_i)\vec{v}_{i-1} + (-c_{i+1}/c_i)\vec{v}_{i+1} + \cdots + (-c_m/c_i)\vec{v}_m.$$

Thus $\vec{v}_i$ is a linear combination of the other vectors on the list.
That is exactly the first definition. □

Video 11.1.3

If a list is not linearly dependent then we will declare it to be **linearly independent**.

It is convenient to write down an explicit definition for this:

Definition ('Linear independence'.)

Fix some Euclidean space $\mathbb{R}^n$.
Consider some finite list of vectors in this $\mathbb{R}^n$

$$\vec{v}_1, \dots, \vec{v}_m \in \mathbb{R}^n.$$

This list of vectors is said to be **linearly independent** if the only $c_1, \dots, c_m$ solving the equation

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_m\vec{v}_m$$

are

$$c_1 = c_2 = \cdots = c_m = 0.$$

Example:

Consider the following list of vectors from $\mathbb{R}^3$:

$$(1,0,0), (0,1,0), (0,0,1).$$ $\to$ standard bases of $\mathbb{R}^3$

Show that these vectors are linearly independent.

Solution:

We'll work directly from the definition.

To show this list is linearly independent, we need to show that the equation

$$c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (0,0,0)$$

only has one solution:

$$c_1 = c_2 = c_3 = 0.$$

Solution (continued):

So let's go ahead and solve this equation in the way we have learnt.

Calculating the LHS we get the equation:

$$(c_1, c_2, c_3) = (0,0,0).$$

Equating corresponding entries, this is a system of three equations:

$$c_1 = 0, c_2 = 0, c_3 = 0.$$

Thus the list is linearly independent.

$\square$

Let's try a less trivial example.

Exercise:

Consider the following list of vectors from $\mathbb{R}^4$:

$$(1,0,-1,0), (1,2,3,1), (1,-2,-5,-1).$$

Is this set of vectors linearly independent or linearly dependent?

Solution:

Again, there is nothing particularly difficult in this problem.

We'll just work directly from the definition and solve linear equations in the way we have learnt.

Solution (continued):

To understand whether this list is linearly independent or not, we need to understand the set of solutions to the following equation $(\star)$:

$$c_1(1,0,-1,0) + c_2(1,2,3,1) + c_3(1,-2,-5,-1) = (0,0,0,0).$$

The thing we need to understand is:

Obviously we can solve this equation by setting

$$c_1 = c_2 = c_3 = 0.$$

But are there any other solutions??

We just solve this equation in the normal way and see.

Solution (continued):

So we compute the LHS into a single vector to get:

$$(c_1 + c_2 + c_3, 2c_2 - 2c_3, -c_1 + 3c_2 - 5c_3, c_2 - c_3) = (0,0,0,0).$$

Equating corresponding entries gives the following system of four equations:

$$\begin{cases} c_1 + c_2 + c_3 & = & 0 \\ 0c_1 + 2c_2 - 2c_3 & = & 0 \\ -c_1 + 3c_2 - 5c_3 & = & 0 \\ 0c_1 + c_2 - c_3 & = & 0 \end{cases}.$$

To solve as usual we start by writing down the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ -1 & 3 & -5 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right].$$

Solution (continued):

Then we perform Gaussian elimination as normal:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ -1 & 3 & -5 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 4 & -4 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

↳ free parameter

infinitely many sol'n
↓
original lst is
linear dependent

- From this we can see that the set of solutions to this system, and hence the vector equation $(\star)$, will have a free parameter.
- Thus there will be infinitely many solutions.
- In particular there will be a solution besides the trivial one $c_1 = c_2 = c_3 = 0$.
- So the system is ***linearly dependent***.

$\square$

Discussion:

That's all we needed to do to answer the question. But it may be interesting to see the dependence explicitly.

Continuing with Gauss-Jordan, the augmented matrix we end up with is:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

This corresponds to the system of linear equations:

$$\begin{cases} c_1 & & +2c_3 & = & 0 \\ & c_2 & -c_3 & = & 0 \end{cases}$$

So, as usual, we set $c_3 = s$, a free parameter, and solve for the remaining variables.

Discussion (continued):

The set of solutions is:

$$\begin{cases} c_1 & = & -2s \\ c_2 & = & s \\ c_3 & = & s \end{cases}, \qquad s \in \mathbb{R}.$$

Substituting this solution back into equation $(\star)$, for every possible $s \in \mathbb{R}$ we deduce that:

$$(-2s)(1,0,-1,0) + (s)(1,2,3,1) + (s)(1,-2,-5,-1) = (0,0,0,0).$$

You can easily double-check this equation, which is a good idea whenever we have the opportunity!

Also: we could choose an s, like $s = -1$, and can rearrange this to demonstrate an explicit instance of linear dependence according to the first definition:

$$(1,2,3,1) = (2)(1,0,-1,0) + (-1)(1,-2,-5,-1).$$

Video 11.1.4

Exercise:

Consider the following list of 4 vectors from $\mathbb{R}^4$:

$$(1,-1,1,-1),(0,1,0,1),(1,1,0,0),(2,1,1,1).$$

Is this set of vectors linearly independent or linearly dependent?

Must solve

$c_1(1,-1,1,-1)+c_2(0,1,0,1)+c_3(1,1,0,0)+c_4(2,1,1,1)=(0,0,0,0)$ cause we in $\mathbb{R}^4$.

This is equivalent to a homogenous system of 4 linear equations in the variables c_1,c_2,c_3,c_4.

$(c_1+c_3+2c_4,\ -c_1+c_2+c_3+c_4,\ c_1+c_4,\ -c_1+c_2+c_4)=(0,0,0,0)$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ -1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 1 & 0 \end{array}\right] \xrightarrow[R_4 \to R_4+R_1]{R_2 \to R_2+R_1,\ R_3 \to R_3-R_1} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 3 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$c_4=0$

$-c_3-c_4=0$

$\therefore c_3=0$

$c_2+2c_3+3c_4=0$

$\therefore c_2=0$

$c_1+c_3+2c_4=0$

$\therefore c_1=0$

Because $c_1=c_2=c_3=c_4=0$ is the only solution, the given list of vectors is linearly independent.

An algorithm to test linear independence

Video 11.1.5

Let's develop a matrix algorithm to answer the following problem:

The computational problem:

Is a given set of vectors $\{\vec{v}_1,\dots,\vec{v}_k\}\subset\mathbb{R}^n$ linearly independent?

The input data:

- A positive integer n and a positive integer k.
- A list of k vectors in $\mathbb{R}^n$:

$$\begin{array}{ccc} \vec{v}_1 & = & (v_{11},\dots,v_{1n}) \\ & \vdots & \\ \vec{v}_k & = & (v_{k1},\dots,v_{kn}) \end{array}$$

Discussion:

The given set of vectors $\vec{v}_1, \dots, \vec{v}_k$ will be linearly independent if the only solution to the following equation is the trivial solution $c_1 = c_2 = \cdots = c_k = 0$.

$$(0, \dots, 0) = c_1(v_{11}, \dots, v_{1n}) + \cdots + c_k(v_{k1}, \dots, v_{kn}).$$

Infinitely many solutions

Only the trivial solution.

If there are infinitely many solutions $(c_1, \dots, c_k)$ to this equation, then there is a non-trivial solution, so the list is ***linearly dependent***.

If the only solution to this equation is the trivial solution $(c_1, \dots, c_k) = (0, \dots, 0)$ then this list is indeed ***linearly independent***.

Discussion (continued):

So. We need to try and solve this system for $(c_1, \dots, c_k)$:

$$(0, \dots, 0) = c_1(v_{11}, \dots, v_{1n}) + \cdots + c_k(v_{k1}, \dots, v_{kn}).$$

This equation is equivalent to a system of n linear equations in the k variables $c_1, \dots, c_k$.

We calculated the augmented matrix earlier for the spanning set algorithms. The answer is:

$$\left[\begin{array}{ccc|c} v_{11} & \cdots & v_{k1} & 0 \\ \vdots & \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} & 0 \end{array}\right].$$

A different way to write this is: $\left[\begin{array}{ccc|c} \uparrow & & \uparrow & \uparrow \\ \vec{v}_1 & \cdots & \vec{v}_k & \vec{0} \\ \downarrow & & \downarrow & \downarrow \end{array}\right].$

Discussion (continued):

To solve this system we would then do Gaussian elimination to put this matrix into row echelon form. We get a matrix like the following, where $\boldsymbol{M}$ denotes the part to the left of the partition which will be in row echelon form, and the part to the right of the partition is $\mathbf{0}$, all zeroes:

$$[\boldsymbol{M}|\mathbf{0}].$$

There are two possibilities for this homogeneous system:

Possibility #1: If there is a column without leading entries in $\boldsymbol{M}$ then the system has infinitely many solutions, and hence the original list is *linearly dependent*.

Possibility #2: If every column has a leading entry, then the system only has the trivial solution. Thus the system is *linearly independent*.

We can now write down the algorithm to test linear independence:

Step 1

Construct an augmented matrix by making the vectors $\vec{v}_1 \dots \vec{v}_k$ the columns of the matrix. (Note: there is no need to bother writing down the column of zeroes to the right of the partition because we don't use it and we are just describing an algorithm.)

$$\begin{bmatrix} \uparrow & & \uparrow \\ \vec{v}_1 & \cdots & \vec{v}_k \\ \downarrow & & \downarrow \end{bmatrix} = \begin{bmatrix} v_{11} & \cdots & v_{k1} \\ \vdots & \vdots & \vdots \\ v_{1n} & \cdots & v_{kn} \end{bmatrix}.$$

The algorithm (continued):

Step 2

Apply Gaussian elimination to put this matrix into row echelon form $\boldsymbol{M}$.

Step 3

There are two possibilities:

- If in the row echelon form $\boldsymbol{M}$ every column has a leading entry of some row then the list of vectors is **linearly independent**.
- If in the row echelon form $\boldsymbol{M}$ there is a column without a leading entry of any row then the system is **linearly dependent**.

Video 11.1.6

Let's do that example we did earlier but now just apply this algorithm we just developed.

Exercise:

Consider the following list of vectors from $\mathbb{R}^4$:

$$(1,0,-1,0), (1,2,3,1), (1,-2,-5,-1).$$

Is this set of vectors linearly independent or linearly dependent?

Solution:

The first step in the algorithm is to construct a matrix whose columns are the given vectors. We write down:

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -2 \\ -1 & 3 & -5 \\ 0 & 1 & -1 \end{bmatrix}.$$

The second step in the algorithm is to perform Gaussian elimination to put this matrix into row echelon form:

$$\to \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & -4 \\ 0 & 1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The third step in the algorithm is to interpret this matrix. In this case: because there is a column without a leading entry (the third column) we deduce that the list is linearly dependent. (This is the same result we got earlier.) $\square$

Please try an example yourself.

Exercise:

Consider the following list of vectors from $\mathbb{R}^3$:

$$(1,1,1), (1,2,3), (2,0,-1).$$

Is this set of vectors linearly independent or linearly dependent?

Solution:

row echelon form.

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 0 \\ 1 & 3 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

Because every column contains a leading entry of a row, it is linearly independent